

Breakdown of Additive Strain–Doping Energetics in Quasi-Hexagonal C₆₀ Graphullerene

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We identify a previously overlooked nonlinear energetic coupling in covalent molecular networks: when substitutional chemistry and biaxial strain are applied jointly, total energies need not equal the sum of separate strain-only and doping-only increments. In high-throughput screening workflows these channels are still commonly combined additively, implicitly neglecting the cross term. Using first-principles calculations on quasi-hexagonal C₆₀ graphullerene, we quantify this omitted coupling with a matched four-corner total-energy difference $\mathcal{S}$ (meV/atom)—a discrete estimate of the bilinear strain–composition term in an energy expansion, not a separate fitted interaction parameter. On B/N/P tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$) at matched PBE+D3 under a rigid-strain protocol, the linear strain coefficient α does not reliably predict the nonlinear coupling energy $|\mathcal{S}|$ (Fig. 3). Among the three substitutional dopants investigated, phosphorus shows the largest $|\mathcal{S}|$ on periodic cells under the rigid-lattice approximation, consistent with its largest covalent-radius excess relative to carbon. At $n=4$ and +3% biaxial strain, the additive fractional error $\eta = |\mathcal{S}|/|\Delta E_{\text{coupled}}|$ reaches $\sim 8\%$ for P, whereas B and N remain below 1% on the same grid (Eq. (4)). At $n=1$ the rigid-strain upper bound reaches $|\mathcal{S}| = 31.9$ meV/atom ($\sigma_S \lesssim 2$ meV/atom from protocol audits), far exceeding the boron and nitrogen values at matched load. These magnitudes are not equilibrium thermodynamic quantities: structural relaxation releases dopant-induced internal stress and can suppress $|\mathcal{S}|$ by orders of magnitude (periodic $n=1$ P at +3%: -31.9 meV/atom rigid vs. ~ 0.1 $\mu\text{eV}/\text{atom}$ relaxed at matched PBE+D3), so $|\mathcal{S}|$ quantifies omitted coupling under mechanically constrained or kinetically limited load. Such rigid-strain upper bounds remain relevant when epitaxial clamping, substrate adhesion, or ultrafast loading limit ionic relaxation during processing or mechanical loading. These results on quasi-hexagonal C₆₀ graphullerene identify nonlinear strain–dopant coupling as an omitted energetic contribution—when dopant-induced internal stress interacts nonlinearly with external biaxial strain—that can alter relative energetic stability under mechanically constrained loading, providing a practical criterion for when additive composition–strain workflows are expected to fail.

I. INTRODUCTION

Substitutional doping and mechanical strain are widely used strategies for tuning the energetic and electronic properties of covalent molecular networks [1, 2]. While external biaxial strain modifies global bonding geometry, substitutional dopants introduce localized chemical stress and lattice distortion, suggesting that the two perturbations need not be energetically independent—i.e., the total response need not be separable into strain-only and composition-only increments without a nonlinear interaction term. In many high-throughput mapping studies, strain and substitution are nevertheless evaluated in separate density-functional theory workflows and combined additively, in part because exhaustive coupled (ϵ, δ) grids are computationally expensive [1, 3, 4]. Whether this additive approximation remains valid under simultaneous load has not been systematically examined for graphullerene networks. Quasi-hexagonal (qHP) C₆₀ graphullerene [5, 6] is a covalently linked molecular network whose mechanical, electronic, and transport response has been mapped under separate strain or substitutional protocols [2, 4, 6–14]. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [15] ($\sim 0.1\text{--}$

0.2 eV-scale cage distortions relative to isolated C₆₀), so a substitutional defect introduces a *local* bond-distortion and chemical-mismatch field that can interact with a *global* biaxial load through elastic frustration in a pre-strained, soft cage—mechanistically beyond Eshelby linear superposition in a stiff continuum. However, these studies primarily report band edges, transport, optics, or mechanics under either strain or substitution alone, whereas the *total-energy* coupling between simultaneous biaxial load and substitutional chemistry—and its implications for relative energetic stability and dopant-site preference under load—has received little systematic attention. Non-additive strain–doping effects are documented for other two-dimensional hosts [3, 16, 17], often through electronic-transport or gap shifts rather than matched four-corner energetics; configurational and cluster-expansion frameworks for alloys routinely include composition–strain cross terms in effective Hamiltonians [18], but it remains unclear whether analogous nonlinear coupling is numerically significant for soft graphullerene cages and when additive composition–strain workflows fail. On qHP C₆₀, whether dopant-induced local stress couples nonlinearly with externally imposed biaxial strain—and how large the omitted energetic correction is at periodic supercell scale—has not been quantified for

substitutional B/N/P tetramers and $n = 1\text{--}8$ supercells.

Here we ask whether the total-energy landscape $E(\epsilon, \delta)$ for substitutionally doped qHP C_{60} under biaxial strain remains additive, because relative energetic stability under load—not band-edge shifts alone—controls which substitutional configurations are favored in processing and under mechanical load. We focus on qHP C_{60} rather than rigid graphene or planar COF hosts because its mixed pentagon–hexagon topology and intrinsically prestrained fullerene cages [15] form a *soft*, covalently linked network in which local defect fields can interact nonlinearly with a global biaxial load—a regime where cluster-expansion bilinear terms are expected to matter but have not been audited at periodic scale [3, 18]. Substitutional B, N, and P are chosen as a minimal III/V-neighbor set spanning donor, acceptor, and size-mismatched channels ($\sim 7\text{--}30$ pm covalent-radius excess vs. C; Table VII); they are not intended as a complete chemical screen (Al, Si, Ga, As, S, etc. would extend the map but are outside the present matched grids). We hypothesize that dopant-induced local stress couples nonlinearly with externally imposed strain, producing energetic corrections beyond additive composition–strain models; on B/N/P chemistry we therefore test three questions: (i) does additive energetics fail under joint (ϵ, δ) load; (ii) when and how large is the omitted coupling, including under mechanically constrained configurations where ionic relaxation is limited; and (iii) does failure track covalent-radius mismatch or donor/acceptor π -electron nonlinearities rather than linear strain coefficients α alone. First-principles matched four-corner audits quantify the omitted term as a same-functional coupling energy $\mathcal{S}$ [Eq. (2)] (Sec. II C): algebraically equivalent to a four-state interaction energy or discrete mixed second derivative of total energy, but reported here as a *physics-first* audit of additive failure rather than as a new thermodynamic observable. The Taylor bilinear term [Eq. (1)] and cluster-expansion interaction-energy framing place the study in the continuum of composition–strain effective Hamiltonians [18], which is why we target *Physical Review B* rather than a methods-only venue. We find that linear α does not predict $|\mathcal{S}|$ or its sign; phosphorus carries the largest rigid-strain upper bound among the three dopants investigated ($|\mathcal{S}| = 31.9$ meV/atom at $n=1$, +3% biaxial strain), pointing to local chemical-mismatch stress as the primary energetic channel rather than π -DOS or charge-transfer proxies alone. Although demonstrated on qHP C_{60} , the matched four-corner criterion offers a practical test for when additive workflows are likely to fail in other covalent networks with comparable local lattice distortion under external load.

II. COMPUTATIONAL METHODS

A. Electronic structure method

All total-energy calculations use CP2K [19] with Goedecker–Teter–Hutter (GTH) pseudopotentials, the DZVP-MOLOPT-SR-GTH atom-centered basis, and the Gaussian and plane waves (GPW) scheme.

a. Exchange-correlation and dispersion. We use PBE [20] with Grimme DFT-D3 (Becke–Johnson damping) [21] rather than hybrid (e.g., HSE06), meta-GGA (e.g., SCAN, r^2 SCAN), or Koopmans-compliant rVV10 protocols [6] because the production dataset requires matched four-corner total energies on dense (ϵ, δ, n) grids at meV-scale resolution. Central conclusions are *within-functional* energy differences: $\mathcal{S}$ subtracts four total energies computed with identical XC, basis, and numerical settings, so systematic errors in absolute gaps or substitution energies largely cancel (Sec. IV). D3 is retained because the qHP network is covalently welded but retains inter-cage van der Waals gaps in the plane; D3 stabilizes long-range dispersion between neighboring C_{60} units without altering the covalent topology. On the matched-functional tetramer P grid, Grimme-D3 non-additivity contributes only $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom, far below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}|$ (Table V). A full SCAN or r^2 SCAN reranking test on a representative corner is not yet in the dataset (Limitations).

b. Plane-wave mesh and Brillouin-zone sampling. Plane-wave cutoffs are 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (REL_CUTOFF 50 Ry; QS_EPS_DEFAULT $10^{-10} E_h$). Comparisons at fixed n always use the same cutoff. Production supercells contain 60–480 atoms with in-plane lattice constants up to ~ 44 Å at $n=8$; inputs use Γ -only sampling (no explicit k -mesh beyond the zone center), which is standard for such large cells when reporting meV-scale total-energy differences within the same supercell.

c. SCF convergence and four-corner precision. Unless noted otherwise, the SCF threshold is EPS_SCF = $10^{-6} E_h$ in inner and outer orbital-transformation loops (MAX_SCF 300; OUTER_SCF MAX 30). Charged-state and vertical workflows in the Supplemental Material may tighten this to $10^{-7} E_h$ (SI Methods). Because $\mathcal{S}$ is a second-order energy difference built from four independently converged totals [Eq. (2)], its numerical uncertainty is set by *cancellation* among corners computed with identical protocols rather than by the absolute SCF tolerance on a single total energy alone. A partial cutoff audit at $6 \times C_{60}$ N and +3% shifts one corner energy by only ~ 0.057 meV/atom between 350 and 400 Ry—well below the smallest $|\mathcal{S}|$ reported in the $n \leq 4$ core grid (e.g., 2.3 meV/atom for N at $n=1$). If a single corner drifts by δ while the other three remain fixed, $|\mathcal{S}|$ changes by $\mathcal{O}(\delta)$ at most; direct $\mathcal{S}(\text{cutoff})$ and $\mathcal{S}(k)$ curves remain a targeted sensitivity appendix (Limitations).

d. Spin, charge, and mechanical boundary conditions. Substitutional periodic cells are charge-neutral closed-

shell Kohn–Sham calculations (no spin polarization in the production $\mathcal{S}$ grid); Hirshfeld strain-path audits on $n=1$ B/N/P use spin-polarized KS only where odd electron counts require it (Sec. IID 3). Doping is modeled as neutral substitution without compensating background charge; ΔE_{sub} is not a formation enthalpy (Sec. IIB). Biaxial strain scales lattice vectors at fixed fractional coordinates (rigid-strain protocol): a mechanical upper bound relevant when ionic relaxation is kinetically restricted—epitaxial clamping, substrate adhesion, or ultrafast loading—rather than an equilibrium thermodynamic prediction (Sec. IID).

B. Strain and doping models

Biaxial strain is defined as $\epsilon = (a - a_0)/a_0 \times 100\%$, where a (a_0) is the strained (pristine equilibrium) in-plane lattice constant; dopant species $\delta \in \{\text{B, N, P}\}$. The strain–dopant coupling energy $\mathcal{S}$ is defined in Sec. IIC [Eqs. (2)–(3)]. π -projected densities of states are written π -DOS.

Tetramer models ($4 \times \text{C}_{60}$) substitute B, N, or P at nominal ~ 5 at.% on a fixed pseudorandom carbon-site map; configurations are not thermodynamically optimized. Periodic supercells place one substituent per C_{60} (~ 1.67 at.%) on a fixed production site map shared across $n = 1\text{--}8$. Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain) is obtained from rigid PBE+D3 tetramer grids; positive α means tensile strain increases the total energy. The substitution energy difference $\Delta E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is not a thermodynamic formation enthalpy (no elemental chemical potentials). Reported α and $\mathcal{S}$ under the rigid-strain protocol (below) are upper bounds on equilibrium magnitudes when ionic relaxation is allowed (Sec. IID).

a. Rigid vs. relaxed mechanical protocols. *Rigid strain* scales in-plane lattice vectors at fixed fractional atomic coordinates: ionic positions do not relax with ϵ . This models mechanically clamped or ultrafast-loaded configurations (epitaxial clamping, strong substrate adhesion) and yields an upper bound on $|\mathcal{S}|$ when internal stress cannot relax. *Relaxed strain* allows fixed-cell BFGS geometry optimization at each corner before the coupling energy is evaluated on matched coordinates (Table V; Fig. 2(e)); ionic relaxation can reverse or nearly extinguish rigid $\mathcal{S}$ (Sec. IID 2).

C. Strain–dopant coupling energy

a. Energy landscape and Taylor expansion. Joint mechanical and chemical load is specified by biaxial strain ϵ and substitutional species $\delta \in \{\text{B, N, P}\}$. Let $E(\epsilon, \delta)$ denote the total energy of the matched supercell at strain ϵ and dopant label δ (pristine for $\delta=0$). Expanding the energy per atom $e(\epsilon, \delta) = E(\epsilon, \delta)/N_{\text{atoms}}$ about the pristine, zero-strain reference in a continuous com-

position parameter x (holding supercell size fixed) gives, schematically,

$$e(\epsilon, x) \approx e_0 + a\epsilon + bx + c\epsilon x + \mathcal{O}(\epsilon^2, x^2, \epsilon^2 x, \epsilon x^2), \quad (1)$$

where the bilinear term $c\epsilon x$ is the lowest-order *non-additive* coupling between external strain and composition. Sequential strain-only and doping-only workflows estimate $e(\epsilon, x)$ by keeping only the linear terms $a\epsilon + bx$; they omit $c\epsilon x$ and all higher-order cross terms. For finite strain and discrete dopants, the same omitted content is estimated by a four-point finite difference.

b. Four-corner finite difference [Eq. (2)]. We refer to this interaction energy as the *strain–dopant coupling energy* $\mathcal{S}$ (a bilinear coupling metric, not an order parameter for a broken-symmetry phase):

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (2)$$

with $\Delta E(\epsilon, \delta) = [E(\epsilon, \delta) - E(0, \text{pristine})]/N_{\text{atoms}}$ and $\delta=0$ denoting the pristine reference at the same supercell size $n \times \text{C}_{60}$. Algebraically,

$$\mathcal{S}(\epsilon, \delta) = \frac{E(\epsilon, \delta) - E(\epsilon, 0) - E(0, \delta) + E(0, 0)}{N_{\text{atoms}}}, \quad (3)$$

so $\mathcal{S}$ depends only on the four corner totals and is independent of any arbitrary additive offset common to all cells. Only a four-state construction removes the linear strain response, the linear doping shift, and the shared pristine reference in one gauge-fixed estimator; three-point or sequential-difference schemes retain spurious linear contamination. In the limit of small ϵ and a smooth composition path, $\mathcal{S}/(\epsilon \Delta x)$ approaches a discrete mixed derivative $\partial^2 e / \partial \epsilon \partial x$; at the finite +3% loads used here, $\mathcal{S}$ reports the *nonlinear* cross term that additive models drop, not merely an infinitesimal curvature.

c. Schematic energy functional. Eq. (1) can be read as a minimal two-parameter energy functional $e(\epsilon, x) \approx e_{\text{elastic}}(\epsilon) + e_{\text{chem}}(x) + c\epsilon x$, where e_{elastic} and e_{chem} are the strain-only and composition-only increments estimated by separate scans, and $c\epsilon x$ is the omitted bilinear work when a dopant-induced *internal stress field* overlaps nonlinearly with external biaxial load on a soft cage. In linear elastic inclusion theory the cross term vanishes at leading order; qHP C_{60} departs from that limit because intramolecular buckling stores internal stress that external tension can release cooperatively (Discussion).

d. Relation to interaction and formation energies. Algebraically, Eq. (2) has the form of an interaction (coupling) energy, and we claim no novelty in that construction. Its distinct role is *dimensional*: a formation or substitution energy subtracts a single reference state, and a binding or interaction energy measures the coupling of one fixed pair of fragments, whereas $\mathcal{S}$ is a symmetric *double-reference cross term* over the four corners of the two-dimensional (ϵ, δ) control space that is spanned by high-throughput screening. It therefore quantifies a quantity none of the pairwise energies provide: the error

incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans (Sec. III C). Table I contrasts $\mathcal{S}$ with common stability and electronic descriptors on these axes. Consistent with this purely energetic definition, $|\mathcal{S}|$ is *not* reducible to a single geometric ($\Delta\bar{d}$) or charge (Δq) channel (Sec. IV C); it is this irreducibility, rather than any new functional form, that motivates reporting $\mathcal{S}$ as a coupling descriptor for the joint design space.

When $\alpha_\delta = dE/d\epsilon$ denotes the *total*-cell linear strain coefficient (meV per 1% strain) at fixed n , a first-order estimate of the per-atom cross term is $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ at small ϵ under the rigid-strain protocol; tetramer and periodic n use different N_{atoms} and dopant concentrations, so this link is not used quantitatively in the periodic audit (Discussion, argument scope).

e. Physical meaning, sign, and units. Physically, $\mathcal{S}$ measures the energetic cost of *nonlinear* interaction between externally imposed biaxial strain and dopant-induced internal stress: when internal cage distortion and macroscopic load are applied together, the total energy need not equal the sum of separate strain-only and doping-only increments. We report $\mathcal{S}$ in meV per atom so that values are comparable across supercell sizes $n \times \text{C}_{60}$ at fixed one-substituent-per- C_{60} loading (~ 1.67 at.% in periodic cells); the same per-atom unit is used for tetramer (~ 5 at.%) audits with the understanding that concentration differs (Sec. IV D). By convention, $\mathcal{S} < 0$ means the coupled (ϵ, δ) corner is *more stable* than the additive estimate (cooperative or stabilizing coupling under load), whereas $\mathcal{S} > 0$ means the coupled state is *less stable* than additive sequential scans predict (anti-cooperative coupling). Because δ is a discrete chemical label rather than a continuous composition coordinate, $\mathcal{S}(\epsilon, \text{B})$, $\mathcal{S}(\epsilon, \text{N})$, and $\mathcal{S}(\epsilon, \text{P})$ are not required to follow a single smooth surface in δ ; cross-species comparisons are meaningful only on matched site maps at the stated concentrations.

f. Size scaling and uncertainty. $|\mathcal{S}|$ is not constrained to decay linearly with $1/n$: electrostatic dilution, cutoff choice, and cage-local distortion can make the cross term peak at intermediate or small n (Table VIII; Fig. 2). If each corner total energy is converged independently with an uncorrelated per-atom uncertainty σ_e , a conservative propagation estimate for Eq. (3) is $\sigma_{\mathcal{S}} \approx 2\sigma_e$ (four equal contributions in quadrature). In practice, identical numerical settings across corners induce partial cancellation, so realized $\mathcal{S}$ noise is often below this upper bound (Sec. II D). A partial cutoff audit shifts one corner total by only ~ 0.057 meV/atom between 350 and 400 Ry at $6 \times \text{C}_{60}$ N and +3%, implying $|\Delta\mathcal{S}| < 0.2$ meV/atom for matched protocols when only one mesh parameter drifts. We therefore report conservative internal uncertainties of $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom for the $n \leq 4$ production grid (well below the P peaks at 8–32 meV/atom) and treat $|\mathcal{S}| > \sim 2$ meV/atom as above protocol noise for the smallest audited periodic points (e.g., 2.3 meV/atom for N at $n=1$). Periodic $\mathcal{S}$ values in the main text are

taken from the +3% size-scaling grid (Table VIII).

g. Additive failure metric. To quantify how large a failure of additivity is, we report the *additive fractional error*

$$\eta \equiv \frac{|\mathcal{S}|}{|\Delta E_{\text{coupled}}|}, \quad (4)$$

where $\Delta E_{\text{coupled}} = (E(\epsilon, \delta) - E(0, 0))/N_{\text{atoms}}$ is the coupled corner increment from the pristine reference at the same supercell size. At $n=4$ and +3% strain, $\eta \approx 8\%$ for P ($|\mathcal{S}| = 23.7$ meV/atom vs. $|\Delta E_{\text{coupled}}| \approx 283$ meV/atom), whereas B and N remain below 1% because their $|\mathcal{S}|$ is small relative to the coupled shift dominated by substitution and strain-only increments. η is a *ranking* diagnostic for sequential workflows, not a universal thermodynamic significance threshold; it should be read alongside $|\mathcal{S}|$ and the audited distribution in Fig. 2(d).

D. Validation benchmarks

Three protocol checks govern interpretation: (i) ionic relaxation of the tetramer P four-corner grid and periodic $n=1$ P fixed-cell geometry optimization (Table V; Fig. 2(e)); (ii) the partial cutoff audit on $\mathcal{S}$ described above; and (iii) matched-functional rigid vs. relaxed coordinates on the tetramer P grid (Table V). Tetramer corners use BFGS force thresholds 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr at $\epsilon=0$ and +3%, respectively (displacement thresholds 3.0×10^{-3} and 1.5×10^{-3} bohr; Sec. II D 2). Periodic ionic benchmarks are limited to $n=1$ P on the production map; tetramer Table VII supplies the complementary ~ 5 at.% relaxation geometry at matched functional. Size scaling extends to $n=8$ (480 atoms, in-plane lattice ~ 44 Å); explicit two-body periodic-image convergence tests on $|\mathcal{S}|$ were not performed beyond this grid.

The task-group inventory in Table V (superseded exploratory grids in Table ??) lists N_{conv} and main-text linkage; ionic-relaxation energies appear in Table VII; the alternate-placement grid in Table VIII spans the full B/N/P strain path and six modern PBE+D3 pristine references. Three validation benchmarks—ionic-relaxation tetramer $\mathcal{S}_P$ (Table VII), alternate dopant placement (Table VIII), and a cutoff benchmark at $6 \times \text{C}_{60}$ N @ +3% (Table V)—govern interpretation of size-scaling trends (Secs. III C and IV).

Sec. III B reports E_{sub} only when tied to the verified PBE+D3 grids above (no chemical-potential reference; not a formation enthalpy).

1. Alternate-placement benchmark

An independent dopant configuration at the same nominal ~ 5 at.% loading repeats the B/N/P strain grid

TABLE I. Positioning of the synergy order parameter $\mathcal{S}$ [Eq. (2)] against common stability and electronic descriptors. Algebraically $\mathcal{S}$ has the form of an interaction (coupling) energy; the distinction is *dimensional*, i.e. which axes of the joint (ϵ, δ) control space each quantity resolves. “Additive-screening error” denotes the bias incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans; $\mathcal{S}$ equals that bias by construction (Sec. II B). “Double-reference cross term” denotes a symmetric difference of differences over the four (ϵ, δ) corners rather than subtraction of a single reference state or fragment pair. Entries are conceptual scope, not numerical performance; consistent with its energetic definition, $|\mathcal{S}|$ is *not* reducible to a single geometric $(\Delta\bar{d})$ or charge (Δq) channel (Sec. IV C).

Descriptor	Nonlinear (ϵ, δ) coupling	Additive-screening error	Double-reference cross term	Joint strain + doping design
Formation / substitution energy E_{sub}	×	×	×	×
Binding / interaction energy ΔE_{int}	Partial	×	×	Partial
Linear strain coefficient α	×	×	×	Partial
Band gap E_g / work function	×	×	×	×
$\mathcal{S}$ (this work)	✓	✓	✓	✓

TABLE II. Periodic synergy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (2). Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1.

n (C ₆₀ units)	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
1	B	+3	−8.38
1	N	+3	+2.30
1	P	+3	−31.91
2	B	+3	−2.18
2	N	+3	+6.85
2	P	+3	−28.29
4	B	+3	−6.23
4	N	+3	+2.95
4	P	+3	−23.71

under rigid PBE+D3 (Table VI). Main-text quantitative α and $\mathcal{S}_{+3\%}^{\text{tet}}$ use the reference-placement map in Fig. 3(a,b); the alternate seed 137 grid is archived only. At periodic $n=4$, two independent site maps repeat the four-corner Eq. (2) audit; the $|\mathcal{S}|$ ranking at +3% matches the production map in Table VIII.

2. Ionic-relaxation benchmark

Fixed-cell BFGS geometry optimization is performed on the tetramer pristine/ P corners at $\epsilon=0$ and +3% (force thresholds 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr; displacement thresholds 3.0×10^{-3} and 1.5×10^{-3} bohr). A matched-functional rigid PBE+D3 control at the relaxed coordinates quantifies how $\mathcal{S}$ changes upon relaxation (Table V). The same BFGS thresholds are used for the periodic $n=1$ P four-corner relaxation reported in Fig. 2(c).

3. Hirshfeld population along strain

Hirshfeld atomic charges are computed along $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ on periodic $n=1$ B/N/P cells (spin-polarized KS for odd electron counts). They complement geometric $\bar{d}$ in Table VII.

III. RESULTS

Readout map: Fig. 1 introduces strain baselines and electronic/mechanical paths; Fig. 2 quantifies additive failure and the $|\mathcal{S}|$ distribution; Fig. 3 traces microscopic paths into a weakly correlated α - $\mathcal{S}$ map; Fig. 4 links charge, geometry, and mismatch to $|\mathcal{S}|$.

A. Electronic structure under strain and doping

Figure 1(a) introduces the systems: pristine and B/N/P-substituted tetramer cages at $\epsilon=0$, with dopant sites highlighted. Panels (b)–(d) establish the electronic and mechanical strain baselines. (i) The N π -DOS waterfall [Fig. 1(b)] and $E_g(\epsilon)$ paths [Fig. 1(c)] show N pinning near E_F while pristine remains open; B ($E_g \approx 0.03$ eV), N ($E_g \approx -0.14$ eV), and P ($E_g \approx 0.05$ eV) at $\epsilon=0$ sit on those trajectories; panel (d) shows the pristine mechanical $\Delta E(\epsilon)$ path underlying reference α . (ii) π -DOS redistribution along these paths is consistent with the donor/acceptor channels that feed the nonlinear coupling energy [Fig. 4(c)] rather than acting as a standalone spectroscopic catalog. That electronic/mechanical baseline does not rank periodic $|\mathcal{S}|$ [Fig. 3(d)]; geometry/charge path proxies are developed in Figs. 3–4. Supplemental Material gives full π -DOS

TABLE III. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table IV.

Calculation group	System	Count	Protocol
Archived baseline (legacy PBE)	$4 \times C_{60}$	24	Rigid strain; not used for main-text claims
π -DOS and gaps	$4 \times C_{60}$	12	PBE+D3
IPR (polaron dimer)	Dimer	4	Pristine/B; $\epsilon=0, +3\%$
Hopping J	Dimer	4	PBE+D3
Charged polarons ^a	$4 \times C_{60}$	20	UKS; geometry opt. + vertical SP
Periodic size scaling	$n \times C_{60}$	41	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times C_{60}$ N	1	+3%; 400 Ry single-point
Ionic relaxation	$4 \times C_{60}$	4	Fixed-cell GEO; P corners
Ionic relaxation	$1 \times C_{60}$ P	4	Fixed-cell GEO; periodic
Alternate placement	$4 \times C_{60}$	24	Rigid PBE+D3
Periodic placement	$4 \times C_{60}$	16	$n=4$; two independent maps
Matched-functional rigid	$4 \times C_{60}$	4	PBE+D3 at relaxed coordinates
Reference placement	$4 \times C_{60}$	24	Rigid PBE+D3
Hirshfeld population	$1 \times C_{60}$ B/N/P	18	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures.

TABLE IV. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times C_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times C_{60}$	24	Superseded
Exploratory design grid	$4 \times C_{60}$	20	Archive only

stacks and periodic VBM/CBM isosurfaces (N size series; P $n=4$): frontier density remains cage-localized for $n = 2-6$.

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Rigid PBE+D3 α on the reference map is +58.7, -297.5, and +456.0 meV/% for B, N, and P [Fig. 3(d)]—the same site map as the periodic $\mathcal{S}$ audit; the pristine $\Delta E(\epsilon)$ path is shown in Fig. 1(d). Table VII and Fig. 4 link mechanical spread to local geometry: on the response plane, P shows negligible strain-induced stretch ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B at +3%) yet the largest bubble ($|\mathcal{S}|$), and only P crosses the $|\Delta r_{\text{cov}}| \gtrsim 20$ pm band.

C. Non-additive strain–doping coupling

Figure 2 is the central quantitative result: additive strain–doping scans fail for size-mismatched P, N carries a positive cross term at $n=4$, and ionic relaxation releases internal stress under the rigid upper-bound protocol. Across fifteen periodic +3% points (Fig. 2(d)), $|\mathcal{S}|$ spans 2.2–31.9 meV/atom with a median of 7.8 meV/atom—the peak P value at $n=1$ is an extremum, not a typical audit magnitude. Internal protocol audits bound

$\sigma_{\mathcal{S}} \lesssim 2$ meV/atom on the $n \leq 4$ grid (Sec. II C), so the smallest reported points (2.3–3.0 meV/atom for N and B at $n=1$) sit just above numerical noise whereas P at $n=4$ (23.7 meV/atom) and $n=1$ (31.9 meV/atom) are an order of magnitude larger. Panel (c) compares the additive prediction ($\Delta E_{\text{strain}} + \Delta E_{\text{dop}}$) with the coupled four-corner energy at $n=4$; the signed gap is $\mathcal{S}$ and documents failure of additivity for all three dopants (P largest, $\eta \approx 8\%$). At $n = 4$ (Table VIII), $\mathcal{S} = -6.2 \pm 2$ meV/atom (B), $+3.0 \pm 2$ meV/atom (N), and -23.7 ± 2 meV/atom (P), where $\pm$ denotes a conservative protocol upper bound, not experimental error. Within $n \leq 4$, $|\mathcal{S}|$ is not monotonic in n ; we report protocol values without extrapolating $\mathcal{S}_{\infty}$. Rows with $n \geq 6$ (350 Ry) appear in Supplemental Material Table S1. Negative $\mathcal{S}$ (B, P at $n=4$) means coupled tension stabilizes the doped cell relative to the additive estimate; positive $\mathcal{S}$ (N) means sequential screens overestimate stability under load. Ionic relaxation of periodic $n=1$ P releases cage misfit stress and lowers $|\mathcal{S}|$ [Fig. 2(e)], supporting the upper-bound interpretation rather than refuting non-additive coupling. Table V provides the matched-functional tetramer relaxation benchmark ($\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ vs. $\mathcal{S}_{\text{relaxed}} = -2.28$ meV/atom; sign not preserved).

Figure 3 encodes microscopic strain paths feeding the weak α – $\mathcal{S}$ correlation: $q(\epsilon)$ and $\bar{d} \pm \sigma(\epsilon)$ [panels (a,b)] show that charge swing and bond stretch do not track $|\mathcal{S}|$; panel (c) traces local disorder $\sigma(\bar{d})(\epsilon)$, with B and N broadening under tension while P remains near the

TABLE V. Ionic-relaxation benchmark for the tetramer P -substitution synergy grid (four fixed-cell geometry-optimization corners: pristine and P at $\epsilon=0$ and $+3\%$, same biaxial cell). Relaxed corners and the matched-functional ratio use PBE+D3 ($\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$; sign not preserved). $\mathcal{S}$ follows Eq. (2) with $N=240$ atoms.

Quantity	Rigid (PBE+D3 matched)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_h)$	—	−1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_h)$	—	−1363.9550
$E(P, \epsilon=0) (E_h)$	—	−1374.4415
$E(P, \epsilon=3\%) (E_h)$	—	−1374.4615
$\mathcal{S}$ (meV/atom)	+1.19	−2.28
Sign of $\mathcal{S}$ vs. rigid	+	− (not preserved)

TABLE VI. Archived alternate-placement tetramer grid (PBE+D3 rigid; seed 137). Not used for main-text figures or quantitative claims; retained for reproducibility of the placement-sensitivity audit. Reference-map α and $\mathcal{S}_{+3\%}^{\text{tet}}$ appear in Fig. 3(a,b). Periodic $\mathcal{S}(n)$ is in Table VIII.

Dopant	Placement	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)
B	alternate	+21.4	−2.6
N	alternate	+45.4	+12.6
P	alternate	+989.6	+41.4

$\epsilon=0$ baseline; the endpoint map [panel (d)] places N in the high- $|\alpha|$ /low- $|\mathcal{S}|$ region and P in the high- $|\mathcal{S}|$ region (Pearson $r \approx +0.71$, $R^2 \approx 0.50$, Spearman $\rho \approx 0.50$ over three dopants—not a reliable rank predictor).

ergies under biaxial strain depending on dopant–host size mismatch [17]; spin-resolved impurity engineering on qHP graphullerene instead targets half-semiconductor gaps [10]; we instead quantify the omitted additive cross term $\mathcal{S}$ in total energy.

IV. DISCUSSION

A. Literature context

Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [2–4, 6, 7, 9–15, 17, 22, 23], the distinction here is a periodic *total-energy* audit of joint biaxial (ϵ, δ) load [Eq. (2)] (Table VIII). In alloy and configurational thermodynamics, composition–strain cross terms enter cluster expansions and effective Hamiltonians as interaction energies rather than as separate observables [18]; elastic inclusion theory likewise superposes defect and external strain fields in the linear regime. Our $\mathcal{S}$ is the same mathematical object—a four-corner finite difference of total energy—applied to a soft molecular network where second-order defect–elastic work can exceed linear $\alpha \times \epsilon$ scaling. Foundational DFT surveys of qHP C₆₀ monolayers emphasize anisotropic elastic moduli and optical gaps [7] rather than matched four-corner total-energy grids under joint substitutional and mechanical load. Strain-engineering roadmaps for generic two-dimensional materials [1] likewise foreground band-edge and mobility levers under separate mechanical or chemical protocols. Graphendofullerene derivatives report Curie temperatures tunable through *either* biaxial strain or electrostatic doping applied in isolation [13], reinforcing that joint (ϵ, δ) total-energy corners are not implicit in sequential protocol designs. In other 2D hosts, substitutional defects can raise or lower formation en-

B. Origins of α – $\mathcal{S}$ weak correlation

Substitutional B, N, or P can be viewed as embedding a *localized internal strain field* ϵ_{int} into an already distorted qHP cage baseline [15]. Table VII quantifies the chemical-mismatch scale by covalent-radius excess Δr_{cov} (+7, −6, and +30 pm for B, N, and P vs. C) and by mean dopant–C distance $\bar{d}$ at $\epsilon=0$. External biaxial strain ϵ_{ext} is a uniform macroscopic load on the periodic cell. Sequential high-throughput models implicitly assume that responses to ϵ_{int} and ϵ_{ext} superpose linearly; $\mathcal{S}$ is the total-energy evidence that this superposition fails when both fields are present. Formally, $\mathcal{S}$ is the finite-difference estimate of the bilinear term in a Taylor expansion of the energy per atom [Eq. (1)]: a discrete mixed second derivative of total energy with respect to strain and composition at finite load, not a separate fitted interaction parameter. In the Eshelby picture, a small substitutional defect in a linear elastic continuum produces a strain field that superposes with external load; qHP C₆₀ falls outside this linear elastic regime because the cage is *soft* and pre-strained [15], so second-order defect–elastic work can exceed the increment predicted by linear $\alpha \times \epsilon$ scaling alone.

N introduces excess π electrons that strengthen bonding under tensile strain (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α). Quantitatively, the $\epsilon=0$ gap ordering ($E_g^{\text{N}} < E_g^{\text{B}} < E_g^{\text{P}}$ within PBE;

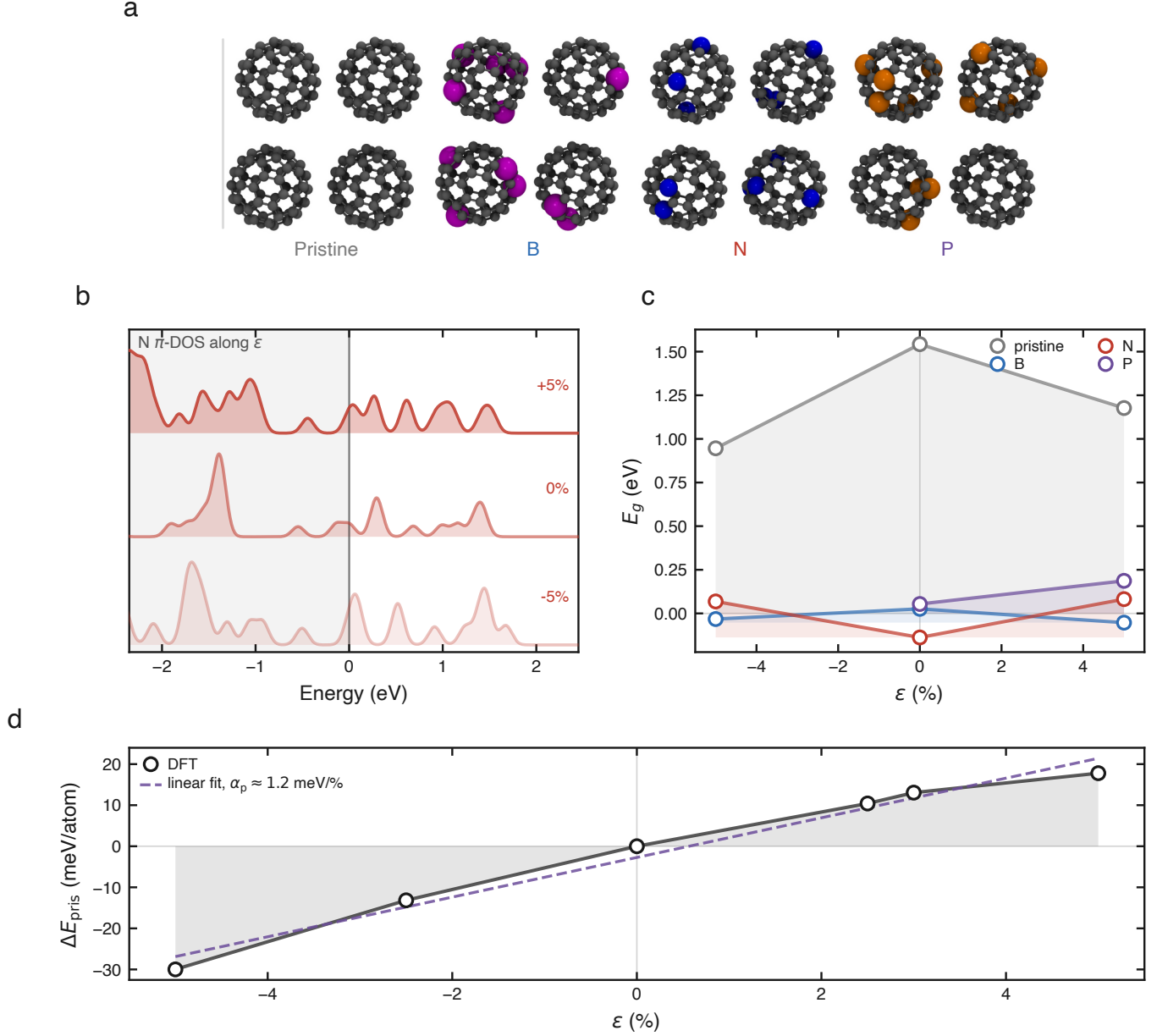


FIG. 1. Process baselines for electronic and mechanical strain response (PBE+D3 unless noted). (a) Ball-and-stick tetramer cages at $\epsilon=0$ (VMD). (b) N π -DOS waterfall at $\epsilon = -5, 0, +5\%$. (c) $E_g(\epsilon)$ trajectories for pristine and B/N/P. (d) Pristine mechanical path $\Delta E(\epsilon)=E(\epsilon)-E(0)$ with a linear guide (reference-map α).

Sec. III A) and the N π -DOS waterfall in Fig. 1(b) align with donor vs. acceptor π -electron counting, while $E_g(\epsilon)$ [Fig. 1(c)] and the α - $\mathcal{S}$ endpoint map [Fig. 3(d)] isolate the reference-map α *sign split* (N negative; B, P positive). α - $\mathcal{S}$ *weak correlation* does not assert anticorrelation between $|\alpha|$ and $|\mathcal{S}|$: N has large $|\alpha|$ but small periodic $|\mathcal{S}|$ at $n=4$, whereas P carries both large $|\alpha|$ and the largest periodic $|\mathcal{S}|$ [Fig. 3(d)]. The claim is that linear strain slopes from a one-parameter scan cannot quantitatively predict four-corner cross-term magnitude or sign on the production site map.

The tetramer linear coefficient α and the periodic cross

term $\mathcal{S}$ probe different energy partitions: α is the slope of total-cell energy under uniform lattice expansion at $\sim 5\%$ multi-site doping, whereas $\mathcal{S}$ is the omitted four-corner cross term per atom at ~ 1.67 at.% under periodic boundary conditions. They therefore need not share sign even for the same element—a negative α (tension lowers total energy in a strain-only scan) can coexist with positive $\mathcal{S}$ (the coupled corner is less stable than an additive model predicts), as for N at $n=4$. Any N sign flip between $n=6$ and $n=8$ lies outside the matched-cutoff $n \leq 4$ core grid and is excluded from design rules.

TABLE VII. Local dopant–cage structure from archived *tetramer* XYZ ($\sim 5\%$ nominal doping; qualitative proxy for mismatch trends). Mean nearest-neighbor C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm. Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$ tracks the uniform nearest-neighbor stretch component. Distances are DFT-derived from archived tetramer XYZ; no experimental bond errors are quoted—comparison is internal across dopants and with qHP literature on intrinsic cage strain [7, 15].

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon = +3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

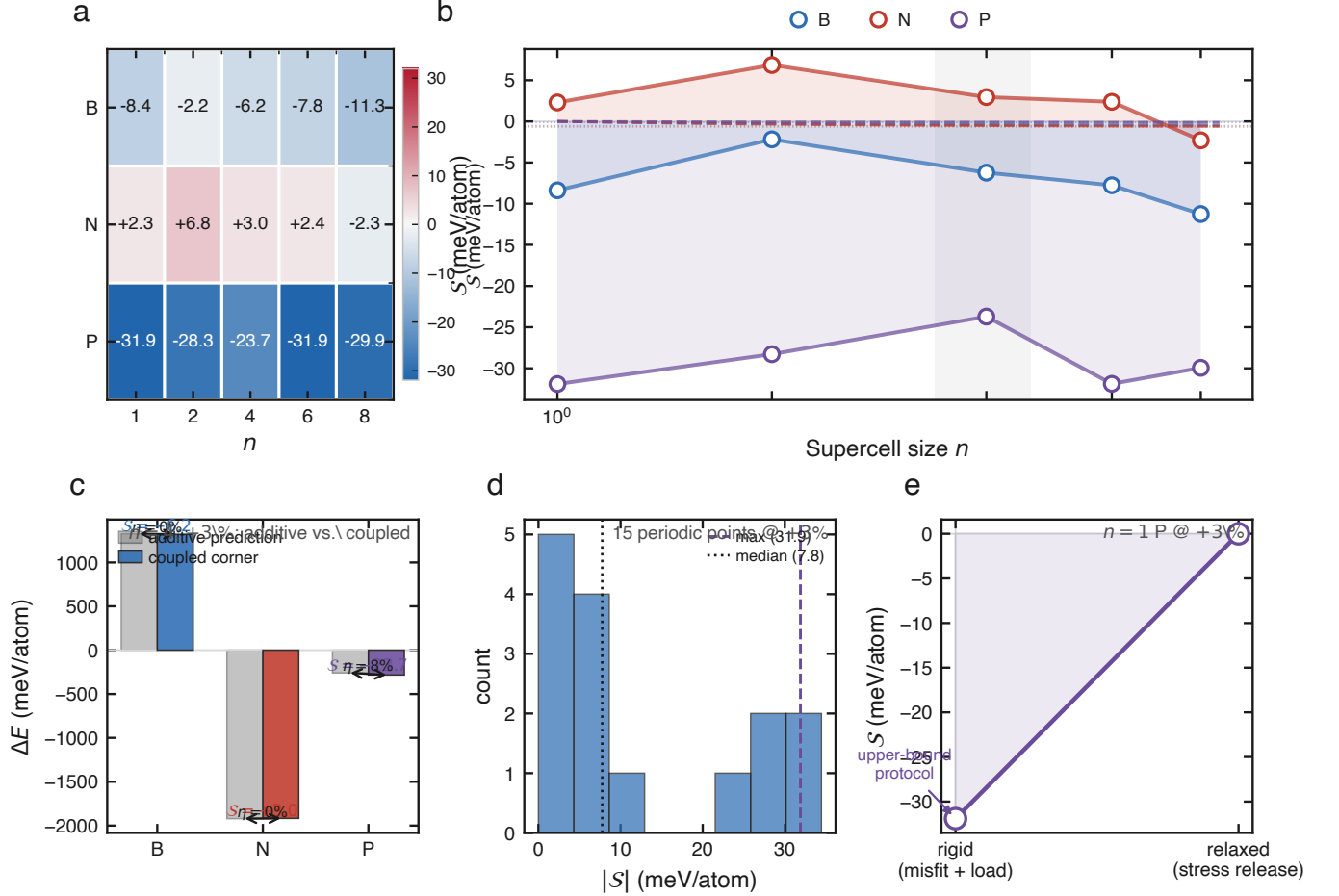


FIG. 2. Non-additive coupling audit (structure insets at key points). (a) Heatmap $S(n, \delta)$ for $n=1-8$. (b) $S(n)$ DFT points with continuous $S_\infty + A/n$ fits (dashed) and provisional S_∞ guides. (c) At $n=4$ and $+3\%$, additive prediction vs. coupled four-corner ΔE (bars); arrows label S and additive fractional error η (Eq. (4)). (d) Distribution of $|S|$ over fifteen periodic $+3\%$ points (median and max marked). (e) Rigid upper-bound vs. relaxed path for $n=1$ P at $+3\%$ (stress release lowers $|S|$).

C. Geometric vs. electronic coupling mechanisms

For *size-mismatched* P, the dopant introduces compressive local misfit in the cage ($\Delta r_{\text{cov}} = +30$ pm) while tensile ϵ_{ext} partially relieves that misfit without proportionally stretching nearest-neighbor bonds ($\Delta\bar{d} \approx 3 \times 10^{-5}$ Å at $+3\%$; Table VII). A radius-only correlation would predict monotonic growth of $|S|$ with Δr_{cov} , but N (-6 pm) yields positive S at $n=4$ while B ($+7$ pm) re-

mains in the few-meV regime, so covalent-radius excess alone cannot rank the three dopants. Figure 4 summarizes the causal chain we infer from the matched audits: covalent-radius mismatch $\rightarrow$ localized internal stress and cage buckling $\rightarrow$ strain-dependent charge redistribution $\rightarrow$ nonlinear total-energy cross term S , with elastic frustration as the decisive channel for P. Three concurrent mechanisms explain the largest $|S|$ without reducing the argument to step one of that chain: (i) *local bond stretch-*

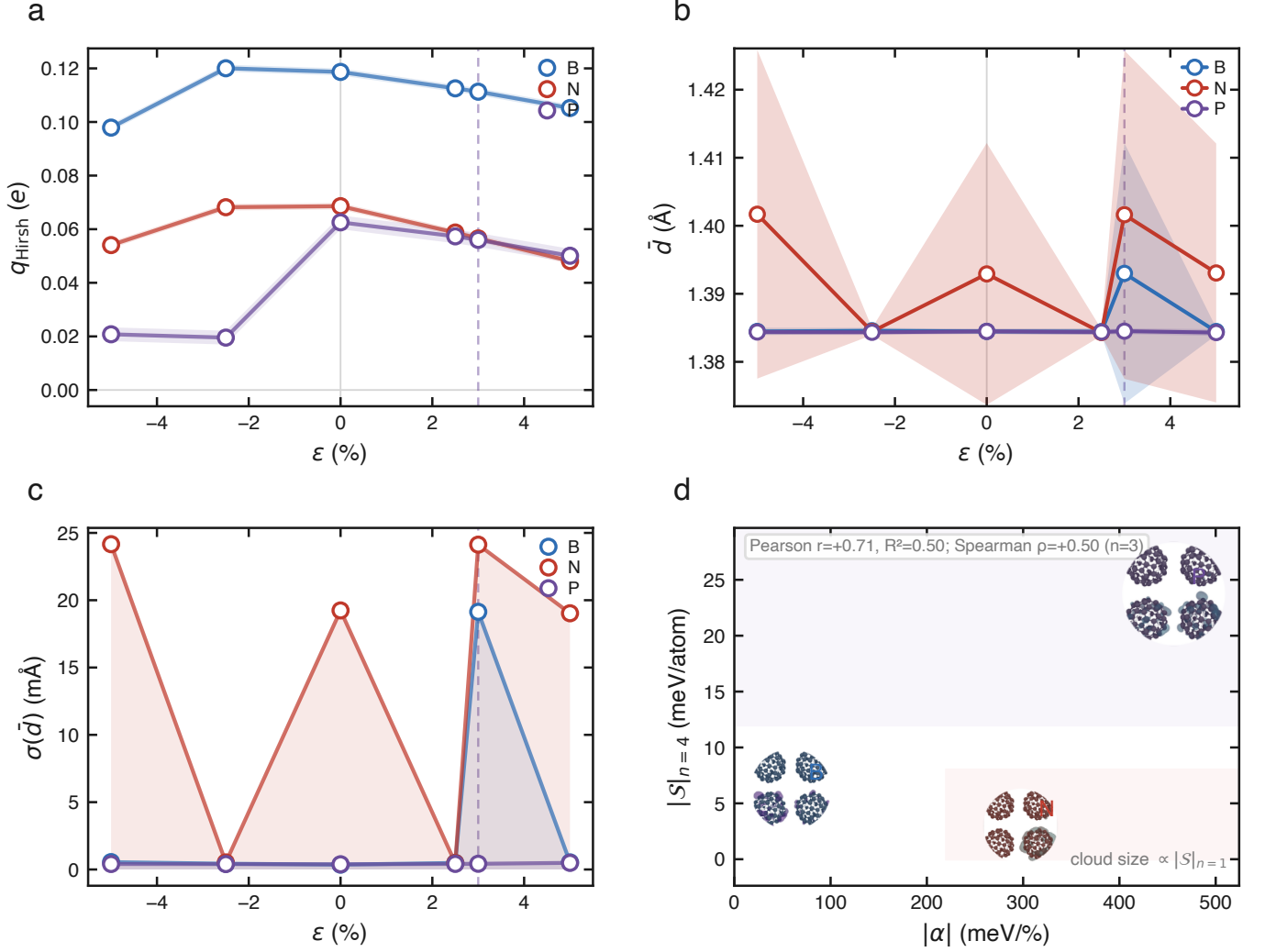


FIG. 3. Microscopic strain paths feeding the α - S map ($n=3$ dopants only). (a) Hirshfeld $q(\epsilon)$ for B/N/P (six-point paths; dotted guide at $+3\%$). (b) Nearest-neighbor $\bar{d}(\epsilon)$ with $\pm\sigma$ bands. (c) Local disorder path $\sigma(\bar{d})(\epsilon)$. (d) Reference-map $|\alpha|$ vs. periodic $|S|_{n=4}$ (marker area $\propto |S|_{n=1}$); Pearson r , R^2 , and Spearman ρ annotated ($n=3$).

ing—P shows the smallest $\Delta\bar{d}$ and $\sigma(\bar{d})$ at $+3\%$, so the cross term is not driven by uniform nearest-neighbor extension; (ii) *charge redistribution*—Hirshfeld Δq on P is moderate ($\sim 11\%$ fractional $|\Delta q/q|$) and does not exceed N's $\sim 18\%$, so charge swing alone does not rank $|S|$; (iii) *elastic frustration*—pre-existing cage buckling [15] plus $+30$ pm misfit stores internal stress that external tension can release cooperatively, producing negative S at $n=4$ (-23.7 meV/atom; Table VIII). The resulting negative S (cooperative stabilization relative to the additive estimate) is consistent with nonlinear defect-elastic coupling: external strain couples to *pre-existing cage buckling* rather than to uniform bond stretching. P carries the largest covalent-radius excess and pre-existing cage distortion at $\epsilon=0$, yet shows the smallest strain-induced change in mean nearest C distance and the smallest site-to-site spread at $+3\%$ ($\sigma(\bar{d}) \approx 4 \times 10^{-4}$ Å vs. $\sim 2 \times 10^{-2}$ Å for B/N; Table VII), yielding the largest $|S|$ at $n=4$. For

B and N, $|\Delta r_{\text{cov}}| \lesssim 10$ pm and $\Delta\bar{d}$ tracks ϵ_{ext} more uniformly ($\sim 8.5 \times 10^{-3}$ Å at $+3\%$), so S is set primarily by electronic nonlinearity—donor vs. acceptor π -electron counting and Fermi-level pinning—rather than by large-amplitude cage reshaping. At $n=4$ the coupled correction for P exceeds the rigid per-atom strain increment from tetramer α by about two orders of magnitude ($|S| \gg \alpha \times 3\%/N_{\text{atoms}}$), even though $|\alpha|$ ranks P as the weakest strain responder on the tetramer grid.

For the matched-functional tetramer P rigid grid, Grimme D3 dispersion energies at the four corners give $S_{\text{vdW}} \approx +0.33$ meV/atom—orders of magnitude below $|S_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.2$ meV/atom (Table V)—so vdW non-additivity does not drive the reported cross terms. On the periodic $n=1$ P grid, Hirshfeld charge on P decreases from $+0.063e$ at $\epsilon=0$ to $+0.056e$ at $+3\%$ while tetramer $\Delta\bar{d}$ for P remains below 10^{-4} Å at $+3\%$ load (Table VII). B and N Hirshfeld charges shift from

TABLE VIII. Periodic synergy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (2). Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1.

n (C ₆₀ units)	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
1	B	+3	-8.38
1	N	+3	+2.30
1	P	+3	-31.91
2	B	+3	-2.18
2	N	+3	+6.85
2	P	+3	-28.29
4	B	+3	-6.23
4	N	+3	+2.95
4	P	+3	-23.71

+0.119 e and +0.069 e at $\epsilon=0$ to +0.111 e and +0.057 e at +3% ($\sim 6\%$ and $\sim 18\%$ fractional $|\Delta q/q|$, vs. $\sim 11\%$ for P). Figure 4(a) makes the ranking failure explicit on path endpoints: large $|\Delta \bar{d}|$ (B) and large fractional $|\Delta q/q|$ (N) alone do not enlarge $|\mathcal{S}|$, whereas P combines moderate Δq with vanishing $\Delta \bar{d}$ and the largest area—geometric pre-strain remains necessary for the P channel [Fig. 4(b,d)]; linked normalized $q/\bar{d}$ trajectories [Fig. 4(c)] trace the charge and bond paths that accompany, but do not linearly predict, the nonlinear coupling energy audited in Fig. 2.

Together, these observations suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α ; *structural* local-distortion coupling to external strain (P); and *finite-size* electrostatic dilution (N at large n), with cage-localized VBM/CBM density on periodic cells (Supplemental Material) as a spatial signature of the third channel.

D. General design rules for covalent networks

Using $|\Delta r_{\text{cov}}|$ from Table VII as a single mismatch coordinate, periodic $|\mathcal{S}|$ spans 2.3–8.4 meV/atom for B and N ($|\Delta r_{\text{cov}}| \lesssim 10$ pm) but reaches 31.9 meV/atom for P ($|\Delta r_{\text{cov}}|=30$ pm; Table VIII). N breaks a naive “ $|\mathcal{S}| \propto (\Delta r)^2$ ” law at comparable mismatch to B because its cross term is electronic (positive $\mathcal{S}$ at $n=4$) rather than geometric. Table VII together with Fig. 4(b) supports a *two-regime* design rule: $|\Delta r_{\text{cov}}| \gtrsim 20$ pm flags the geometric-nonlinear regime where coupled audits are mandatory (P); $|\Delta r_{\text{cov}}| \lesssim 15$ pm leaves $|\mathcal{S}|$ at few-meV/atom under +3% load but still permits sign conflicts between α and $\mathcal{S}$ (B, N). These thresholds are calibrated on qHP C₆₀; rigid graphene-like hosts may require higher mismatch before geometric nonlinearity dominates.

At $|\Delta r_{\text{cov}}| \lesssim 10$ pm and +3% biaxial load, $|\mathcal{S}| \lesssim 10$ meV/atom on the $n=1$ audit grid; sequential pipelines may be adequate for coarse ranking but not for close E_{sub} competitors. Pristine tetramers show $|\alpha| \approx 1$ meV/atom, so sub-1% loads keep linear strain increments small; at +3% and above, coupled corners for size-mismatched

dopants are advisable. qHP C₆₀ combines interfullerene covalent links with soft intramolecular cages and intrinsic intramolecular strain [15], amplifying local-distortion coupling relative to rigid 2D lattices; extrapolation to graphene-like hosts requires separate benchmarks. Tetramer α and $\mathcal{S}_{+3\%}^{\text{tet}}$ are not interchangeable with periodic $\mathcal{S}(n)$ because concentration ($\sim 5\%$ vs. ~ 1.67 at.%) and boundary conditions differ; non-additivity is assessed only through Eq. (2) on matched four-corner grids.

At $n=4$ and +3% strain the coupled correction is $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential additive estimates differ by approximately a factor of four in the load-dependent margin under the same PBE+D3 periodic protocol. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. For B at $n=4$, negative $\mathcal{S}$ coexists with positive α ; for N at $n=4$, negative tetramer α and positive $\mathcal{S}$ present a second sign conflict. Gap ordering along the $E_g(\epsilon)$ paths [Fig. 1(c)] does not track $|\mathcal{S}|$, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone.

E. Experimental signatures

Three classes of laboratory observables could test the stress-field picture without resolving total energies directly. **Raman or IR cage modes:** under joint doping and strain, mode shifts should deviate from the sum of doping-only and strain-only shifts when $|\mathcal{S}|$ is large—especially for P, where tensile load couples to pre-existing cage distortion rather than uniform bond stretching (Table VII). **Scanning tunneling spectroscopy:** dopant-site local density of states should evolve nonlinearly with strain when Hirshfeld charge redistributes without proportional $\Delta \bar{d}$ (Sec. IV C); N pinning near E_F vs. gapped B/P provides contrasting signatures. **Thermal or chemical stability:** for $\mathcal{S} < 0$ (B, P at $n=4$), tension should enhance apparent doping stability relative to additive thermochemical estimates; for $\mathcal{S} > 0$ (N at $n=4$), the opposite bias is expected. Quantitative links require matched sample chemistry and strain calibration;

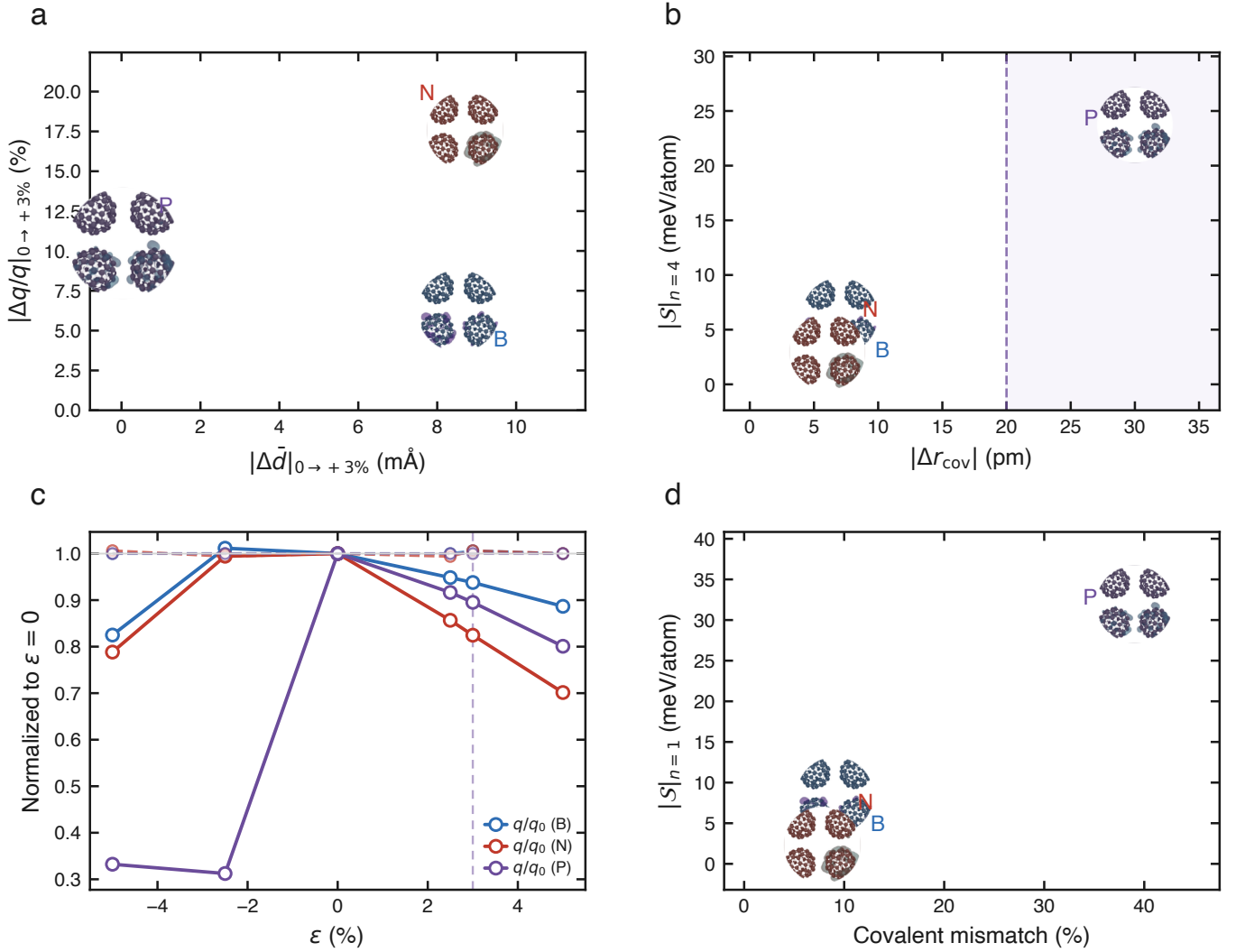


FIG. 4. Mechanism chain and counterexamples (path endpoints, design rule, linked trajectories). (a) Response plane from 0 $\rightarrow$ +3% path endpoints; bubble area $\propto |S|_{n=4}$ (B: large $\Delta \bar{d}$, small $|S|$; N: large $|\Delta q|$, moderate $|S|$). (b) $|\Delta r_{\text{cov}}|$ vs. $|S|_{n=4}$: radius alone does not rank B/N/P (N positive S , B smallest despite +7 pm). (c) Linked normalized paths $q(\epsilon)/q(0)$ (solid) and $\bar{d}(\epsilon)/\bar{d}(0)$ (dashed)—charge and bond paths accompany but do not linearly predict $|S|$. (d) Covalent mismatch (%) vs. $|S|_{n=1}$ (P extremum at largest mismatch).

we quote DFT protocol bounds only.

F. Physical relevance of mechanically constrained configurations

Ionic relaxation can suppress equilibrium $|S|$ by orders of magnitude (periodic $n=1$ P at +3%: -31.9 meV/atom rigid vs. ~ 0.1 μeV /atom relaxed at matched PBE+D3), so rigid-strain S is *not* an equilibrium free-energy difference. Its physical role is to bound omitted coupling when internal stress cannot relax during processing or service: epitaxial films and clamped substrates that pin in-plane lattice constants; strong adhesion that transfers load before cage ions equilibrate; ultrafast or cyclic mechanical loading; and nanoscale flexures or MEMS ele-

ments where boundary stiffness exceeds local cage compliance. In those settings, sequential additive screens can mis-rank substitutional stability by meV/atom margins comparable to the audited $|S|$ distribution (median 7.8 meV/atom across fifteen periodic +3% points; Fig. 2(d)). Equilibrium relaxed audits remain the appropriate benchmark for fully annealed, bulk-like samples; rigid four-corner grids diagnose when kinetics or boundary conditions freeze the internal stress field that drives nonlinear coupling.

G. Alternative explanations and numerical controls

Several non-physical mechanisms could mimic a four-corner cross term if corners were computed inconsis-

tently; we address them explicitly. (i) *Finite-size / electrostatic dilution*: $|\mathcal{S}|$ is not monotonic in n and can peak at $n=1$ (P) or change sign between $n=6$ and $n=8$ for N (Supplemental Material), inconsistent with a single image-sum artifact. (ii) *Charge localization / SCF metastability*: all reported corners show converged SCF totals under identical OT settings; tetramer P metastability at +2.5% is excluded from α fits but does not enter the four-corner $\mathcal{S}$ grids audited here. (iii) *Pulay stress and basis-set incompleteness*: comparisons at fixed n use matched 400 Ry cutoffs; the partial cutoff audit shifts one corner by only ~ 0.057 meV/atom, far below the quoted $|\mathcal{S}|$ for $n \leq 4$. (iv) *Numerical cancellation in Eq. (3)*: because all four totals share the same functional, k-point (Γ), and cell shape, spurious additive offsets cancel; remaining uncertainty is bounded by $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom (Sec. IIC). None of these controls eliminate the P-dominated geometric-nonlinear channel; they set the scale below which we would not interpret $|\mathcal{S}|$ as physics rather than protocol noise.

H. Predictive use of four-corner audits

The matched $\mathcal{S}$ grid is intentionally cheap relative to full ionic relaxation at every (ϵ, δ) point: once four rigid-strain corners converge, $\mathcal{S}$ forecasts whether a sequential strain-plus-doping screen will mis-rank substitutional stability before expensive relaxations or transport calculations are deployed. In practice, $|\mathcal{S}| > \sim 2$ meV/atom at +3% load flags candidates for mandatory coupled re-evaluation; P on the production map exceeds this threshold by an order of magnitude while B and N often remain near the noise floor. This is a *stability-ranking* tool under stated boundary conditions, not a universal replacement for fully relaxed formation energies.

I. Limitations and outlook

Main-text α and $\mathcal{S}$ use PBE+D3 under the rigid-strain protocol (Sec. IIB); ionic relaxation releases internal stress and can reverse or nearly extinguish $\mathcal{S}$ (Table V; periodic $n=1$ P). Although full relaxation suppresses equilibrium $|\mathcal{S}|$, rigid-strain audits bound coupling under epitaxial clamping, substrate adhesion, or ultrafast loading when ionic relaxation is kinetically restricted. Periodic $\mathcal{S}(n)$ magnitudes are configuration-specific on a fixed production site map (two independent $n=4$ maps preserve $P > B > N$; Table VIII). Core claims use matched 400 Ry for $n \leq 4$; $n \geq 6$ production points use 350 Ry. Table VII geometry is a tetramer proxy; the geometric vs. electronic channel assignment rests on $\Delta\vec{d}$, Δr_{cov} , and Hirshfeld Δq rather than a full energy decomposition. Because $\mathcal{S}$ is a same-functional difference, absolute errors in substitution energy do not affect our conclusion of non-additive coupling. Per-corner SCF convergence and the $\sigma_{\mathcal{S}} \approx 2\sigma_e$ propagation bound in

Sec. IIC supply internal uncertainty estimates; we do not quote experimental $\pm$ on bond lengths because Table VII reports DFT-internal geometry only. With only three substitutional labels, Pearson, Spearman, and R^2 statistics on the α - $|\mathcal{S}|$ map are illustrative ($n=3$); bootstrap resampling would not add robustness without additional converged dopant points. Claims are scoped to qHP C₆₀ on fixed site maps; graphene-like rigid lattices or COF bilayers [16] may show smaller cross terms and require separate four-corner benchmarks. Future work includes Mayer/Bader strain-path partitioning, additional III/V substituents on matched maps, direct $\mathcal{S}(\text{cutoff})$, $\mathcal{S}(k)$, and $\mathcal{S}(\text{EPS_SCF})$ sensitivity curves, and a representative SCAN or r²SCAN spot-check on a high- $|\mathcal{S}|$ corner.

V. CONCLUSIONS

Nonlinear strain-dopant energetic coupling—not the symbol $\mathcal{S}$ —is the primary finding: joint load on qHP C₆₀ breaks additive composition-strain workflows when dopant-induced internal stress meets external biaxial strain. The matched four-corner difference $\mathcal{S}$ is the gauge-fixed estimator of that omitted bilinear term [Eqs. (1)–(3)], equivalent to a discrete mixed derivative of total energy rather than a new thermodynamic observable. On qHP C₆₀, donor/acceptor dopants produce an α sign split [Fig. 3(d)] that does not reliably predict $|\mathcal{S}|$, while periodic $|\mathcal{S}|$ reaches 31.9 meV/atom for P at $n=1$ at fixed coordinates—an extremum within a broader 2.2–31.9 meV/atom audit distribution (Fig. 2(d)). Three channels—local bond stretching, charge redistribution, and elastic frustration—jointly explain why P dominates the geometric-nonlinear regime without reducing the argument to covalent radius alone (Sec. IV C). Rigid-strain $\mathcal{S}$ bounds coupling under mechanically constrained loading (Sec. IV F); equilibrium relaxation releases internal stress and can nearly extinguish the cross term. Coupled (ϵ, δ) screening is therefore advisable for $|\Delta r_{\text{cov}}| \gtrsim 20$ pm substituents and +3% loads; four-corner audits can pre-screen such cases before full relaxation (Sec. IV H). Generalization beyond qHP C₆₀ is likely only after matched four-corner audits on each host; the present study documents the phenomenon and audit protocol on a soft covalent molecular network rather than claiming universal constants or materials-general laws.

ACKNOWLEDGMENTS

We acknowledge computational resources and support from Xiamen University.

DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/>

chenxingqiang/graphullerene-strain-engineering, including computational inputs and outputs, tabulated synergy and polaron analyses, and materials to reproduce all figures and tables in the main text.

Ionic-relaxation, Hirshfeld, and size-scaling archives referenced in Tables VI–VIII are included in the public release. A tagged release matching the manuscript revision will be issued upon acceptance.

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